

Arumalla Mohit Krishna

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Education

Stony Brook University, New York, USA

Aug 2025 – May 2027

Master of Science in Computer Science - GPA 3.5/4.0

Relevant Coursework: Distributed Systems, Database Systems, Analysis of Algorithms

Birla Institute of Technology and Science, Pilani

Aug 2019 – July 2023

Bachelor of Engineering in Electronics & Communication - CGPA 8.02/10.0

Relevant Coursework: DSA, OOPS, Operating Systems, Computer Networks, Neural Networks and Fuzzy Logic,

Technical Skills

Languages: Golang, C++, Python, SQL

Backend & APIs: gRPC, RESTful APIs, Microservices, Istio Service Mesh

Distributed Systems: Apache Flink, Kafka, Paxos, RDMA, Change Data Capture (CDC)

Databases: PostgreSQL, Elasticsearch, Redis, RocksDB

Cloud & Infrastructure: AWS (S3, Lambda, Titan Embeddings), Docker, Kubernetes

Tools: Git, Debezium, New Relic, Gorm ORM

Work Experience

Infra.Market, Bangalore, India | Software Engineer 2

Apr 2024 – Aug 2025

- Architected and implemented production AI-driven RAG system with reusable Golang library consumed by multiple teams, serving **100K+ monthly queries** with Elasticsearch vector DB and AWS Titan Embeddings; achieved **28% CTR increase**, **45% zero-results reduction**, improving satisfaction from **3.2 to 4.1** out of **5.0** rating.
- Set up **Apache Flink** real-time data aggregation to synchronize customer data from PostgreSQL to Elasticsearch using CDC with Debezium and service-sent events from Kafka topics, reducing syncing time from **10–15 minutes** of earlier batch processing to sub-second (**500 ms**) latency using **RocksDB** as state backend and **Flink SQL** for complex transformations.
- Optimized backend service handling 50K+ daily queries, reducing memory by **70%** through Elasticsearch client optimization and Golang heap profiling; eliminated Garbage Collection pauses and OOM errors in production.
- Deployed Istio service mesh for automated service-to-service observability, enabling distributed tracing with Envoy proxies and integrating with New Relic for centralized monitoring across **15+ microservices**.

Infra.Market, Chennai, India | Software Engineer 1

Jul 2023 – Mar 2024

- Owned Search and Location microservices, delivering gRPC-based backend services handling **500K+ daily requests** for product discovery and geolocation across distributed cloud infrastructure.
- Implemented automated monitoring with Slack alerts for stale index detection across Elasticsearch, BigQuery, and Airflow, reducing downtime by **60%** and enabling proactive infrastructure maintenance for **5M+ indexed records**.
- Resolved critical PostgreSQL connection leaks in location service using Gorm ORM, reducing connection pool exhaustion by **85%**; validated fixes through concurrent load testing at **10K requests/second** to prevent deadlocks under high traffic.
- Implemented standardized API documentation using Swagger/OpenAPI for all backend services, improving API discoverability and reducing integration time for internal teams by **60%**.

Projects

RDMA-Accelerated Distributed Transactions

Nov 2025 – Ongoing

- Architecting RDMA integration for distributed transaction replication targeting **30-50% latency reduction** and **40-60% CPU improvement** over TCP/IP by implementing one-sided RDMA operations for Paxos-based log shipping.
- Designing remote memory layout and RDMA connection management with both two-sided verbs and one-sided operations for commit notifications, researching Motor (OSDI '24), FaRM, and DrTM for low-latency protocols.

Elasticsearch Configuration Service | GitHub

Apr 2025 – Jul 2025

- Engineered platform abstraction layer with self-serve RESTful APIs in Golang for Elasticsearch operations, enabling automated index lifecycle management, schema versioning, and query optimization, reducing setup time by **70%** with infrastructure-as-code.

Achievements

- Achieved **1900+ contest rating** on LeetCode (Knight badge, top 5-10%) with **500+ DSA problems** solved across competitive programming platforms.
- Secured **3rd place** in Infra.Market Hackathon by developing a GenAI-powered knowledge assistant that indexed Confluence pages, Jira tickets, and Figma files for quick retrieval.